Algorithm

Calculate Carbon Emission and Provide Environmental Insights

1. **Start**
2. **Define Constants and Arrays:**
    - Set the total number of activities as 5.
    - Create an array of activity names (e.g., "Car travel", "Electricity usage", etc.).
    - Create an array of predefined CO2 emissions for each activity.
    - Create an empty array for user inputs.
3. **Display Instructions:**
    - Show a welcome message and instructions for entering activity data.
4. **Input User Data:**
    - For each activity, ask the user to input the number of units (e.g., kilometers, hours, etc.).
    - Check if the input is valid (numeric); if not, show an error and stop the program.
5. **Calculate Total Emissions:**
    - Multiply each activity's units by its CO2 emission factor and sum them up.
6. **Determine Air Quality Index (AQI):**
    - Use the total emissions to calculate AQI based on predefined ranges.
7. **Calculate Climate Impact:**
    - Compute a "climate change impact factor" using a logarithmic formula.
8. **Estimate Offsets Needed:**
    - Calculate how many trees or solar panels are needed to offset the emissions.
9. **Show Results:**
    - Display activity-wise emissions, total emissions, AQI, climate impact factor, and offset suggestions (trees and solar panels).
10. **Display Bar Chart:**

- For each activity, calculate its proportional emissions and display a bar chart.

11. **Show Menu for Further Options:**
    - Present a menu with the following options:
        - View detailed climate report.
        - Compare car vs. bus emissions.
        - Get carbon offset suggestions.
        - Learn environmental tips.
        - View all options.
        - Predict future emissions.
12. **Handle User Choice:**
    - Based on the selected option, perform the corresponding action and display the results.
    - Repeat until the user decides to exit.
13. **Save Report to File:**
    - Save the climate impact report and other results in a text file.
14. **Exit Program:**
    - End the program after saving or user exits.
15. **Stop**